Yingrui Hou

Simulation & Model Validation | Digital Twin Applications

Clermont-Ferrand, France | yingrui.hou@outlook.com | +33(0)785396332
https://yingrui-hou.github.io/PersonalPage/

Profile

Physics PhD with postdoctoral experience connecting experimental measurements, simulation and calibrated models. Builds Python/C++/Geant4 workflows to characterise system response, correct mismatch and validate predictions under uncertainty. Brings measurement-to-model experience transferable to digital twin development, simulation engineering and industrial R&D, with research project coordination, technical presentations and international collaboration.

Core skills

- Simulation: Geant4/C++, detector geometry, interaction stepping, response logic, event-level outputs and measured response maps.
- Validation & calibration: simulation-to-measurement alignment, control samples, bias correction, uncertainty analysis and before/after performance comparisons.
- Signals & models: time-profile decomposition, likelihood fitting, statistical inference, CatBoost/gradient boosting and data-driven model correction.
- Scientific software: Python, C/C++, ROOT/RooFit, NumPy, pandas, Linux; Shell/YAML/CMake automation, reusable components and shared-code review.

Research experience

Postdoctoral Researcher | 2021–2025

Research in CNRS/CERN environments; LHCb and GRAiNITA collaborations.

- Built and adapted Geant4/C++ detector simulations, implementing geometry, stepping and analysis outputs to study optical and detector response.
- Connected beam-test measurements and position-dependent response maps to simulated energy deposits, evaluating performance against measured rather than ideal response.
- Developed time-resolved signal analysis and ROOT/C++ correction workflows; checked reconstructed energy before and after correction under the studied conditions.
- Investigated detector and reconstruction anomalies through reference-channel comparisons; documented causes and validation findings for calibration and software follow-up.

Doctoral research training | 2016–2021

University of Chinese Academy of Sciences — joint Master and PhD in Particle Physics. Training in quantitative problem formulation, scientific computation, experimental validation and communication of research results.

Education

- Joint Master and PhD in Particle Physics — University of Chinese Academy of Sciences, 2016–2021.
- BSc in Applied Physics — China University of Mining and Technology, 2010–2014.

Selected project evidence

Measurement-informed detector model

Used June 2024 beam-test measurements to derive an effective light-yield map and inject it into Geant4 energy-deposition simulation. Compared measured and idealised response using resolution fits. The non-uniformity contribution to the constant term was below 1% in the first GRAiNITA prototype study. Transfer: measurement-informed model calibration and validation for digital twins.

Signal decomposition and correction

Extended simulated scintillation timing with particle-dependent response and fitted component fractions for reconstructed-energy correction. One representative rotation-based PSD study reduced a resolution term by about 2×. Material comparisons also exposed limits of component separation; the result is specific to the studied configuration.

Model validation leading to corrective action

Compared reconstructed photon data with simulation across benchmark channels and selections. Traced discrepancies to detector regions, energy calibration and a missing-energy software issue later fixed downstream. Transfer: connect model-health diagnostics to engineering investigation, while distinguishing personal diagnosis from team implementation.

Research training applied to engineering

- Adaptability and learning: moved between likelihood inference, CatBoost-based selection and Geant4 detector modelling; combined literature, implementation and data checks to apply methods to unfamiliar problems.
- Research project coordination: organised connected analysis tasks from measurement inputs to validated results, with reusable components and documented assumptions for collaboration review.
- Presentation and international teamwork: presented at GDR-InF 2022 and the 5th FCC/DRD France Workshop in 2025, including a talk on behalf of GRAiNITA; contributed to shared code and technical reviews.

Languages & mobility

Chinese: native | English: C1 | French: B1–B2. Long-distance cycling and travel; open to relocation in Europe.

Project references

- Measurement-informed detector model (https://yingrui-hou.github.io/PersonalPage/index.html#work-4)
- Signal decomposition and correction (https://yingrui-hou.github.io/PersonalPage/index.html#work-5)
- Model validation leading to corrective action (https://yingrui-hou.github.io/PersonalPage/index.html#work-6)